

Network Reliability under Edge Failures

Using **Monte Carlo Simulation** and **graph theory** to study how network structure affects reliability when connections randomly fail.

Abdiaziz Muse & Yijun Li

Motivation: Real-World Networks

Many critical systems can be modelled as networks and their links can fail randomly.

Communication

Internet & telecom networks

Transportation

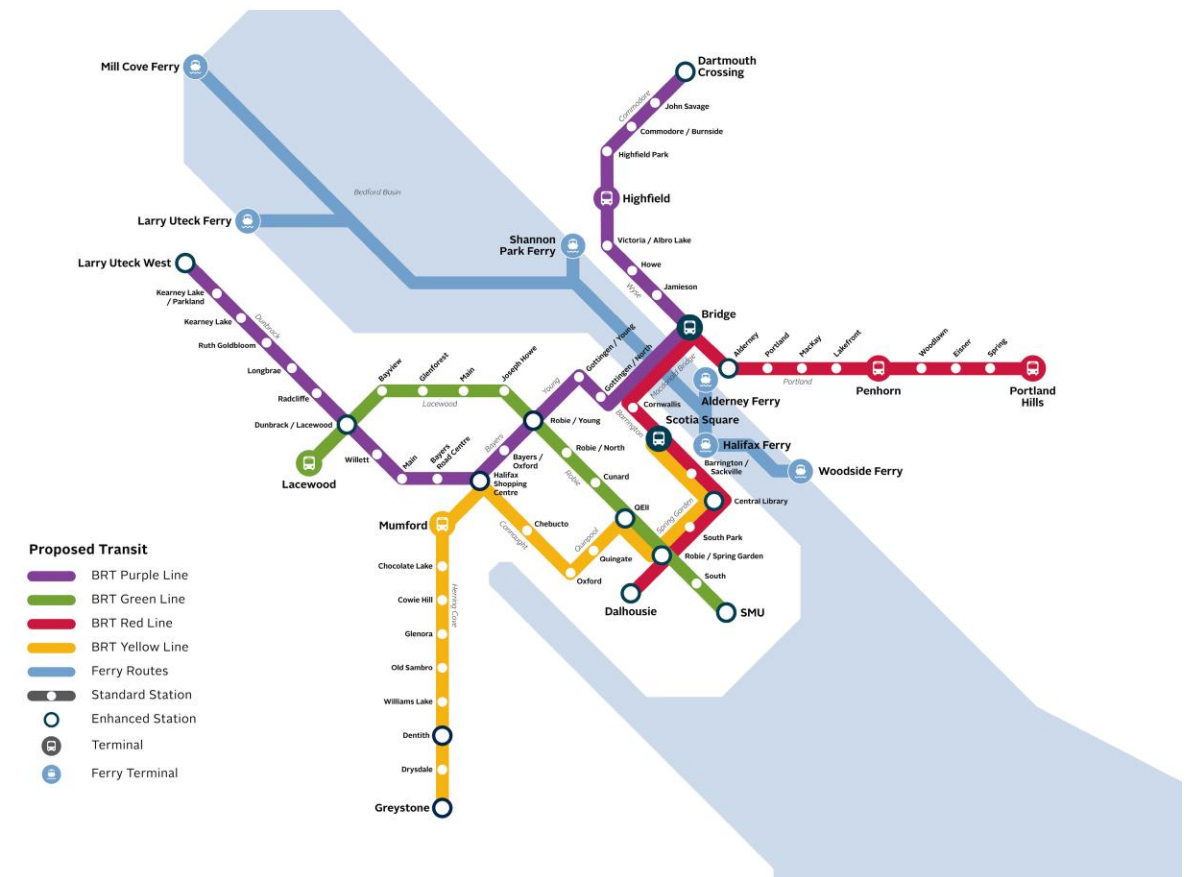
Roads, rails, and flight routes

Power Grids

Electrical distribution systems

Computer Networks

Data centers and servers



Key question: How likely is the network to remain connected when edges fail? This is called network reliability.

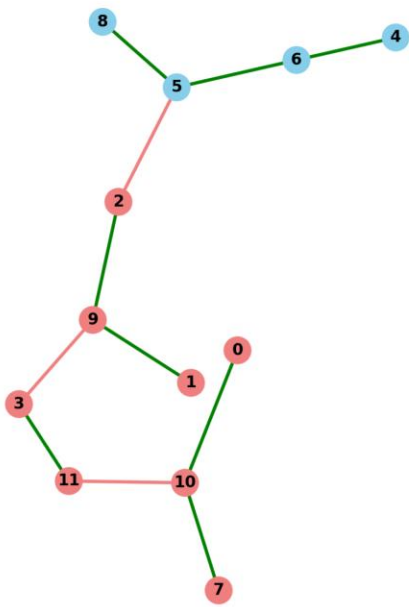
Key Idea: Structure Determines Reliability

Edge-Cut

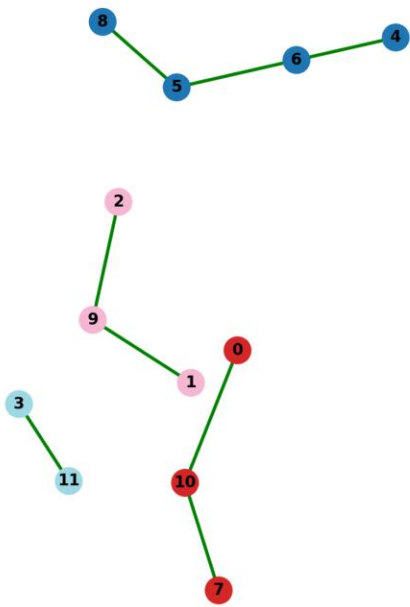
An edge cut is a set of edges whose removal immediately disconnects the graph into two or more components.

Why It Matters

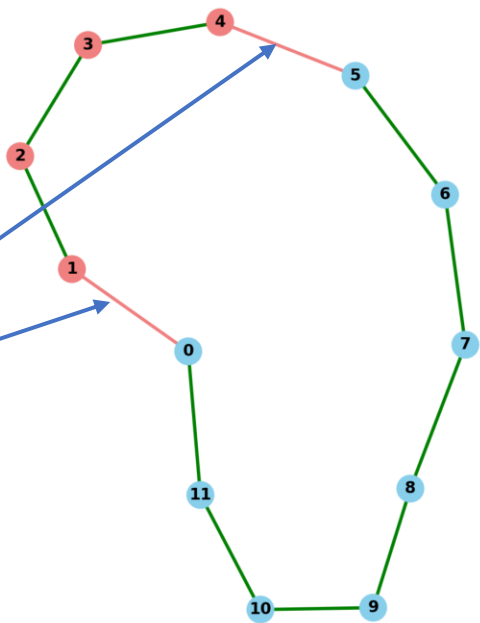
These structures determine **how easily a network can become disconnected**. Our project studies the relationship between graph structure and reliability.



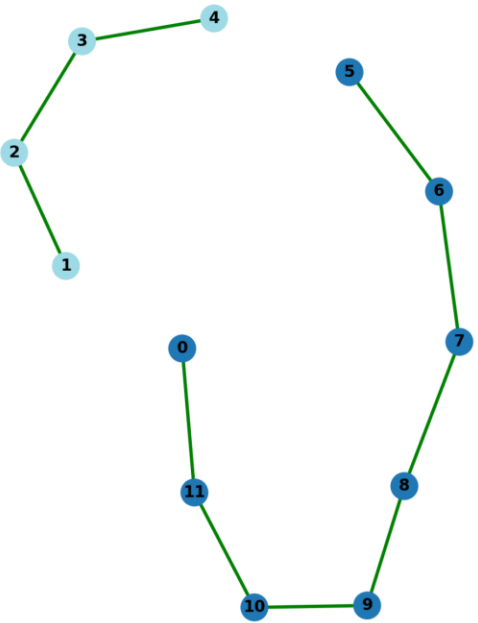
Tree graph



Edge-Cut



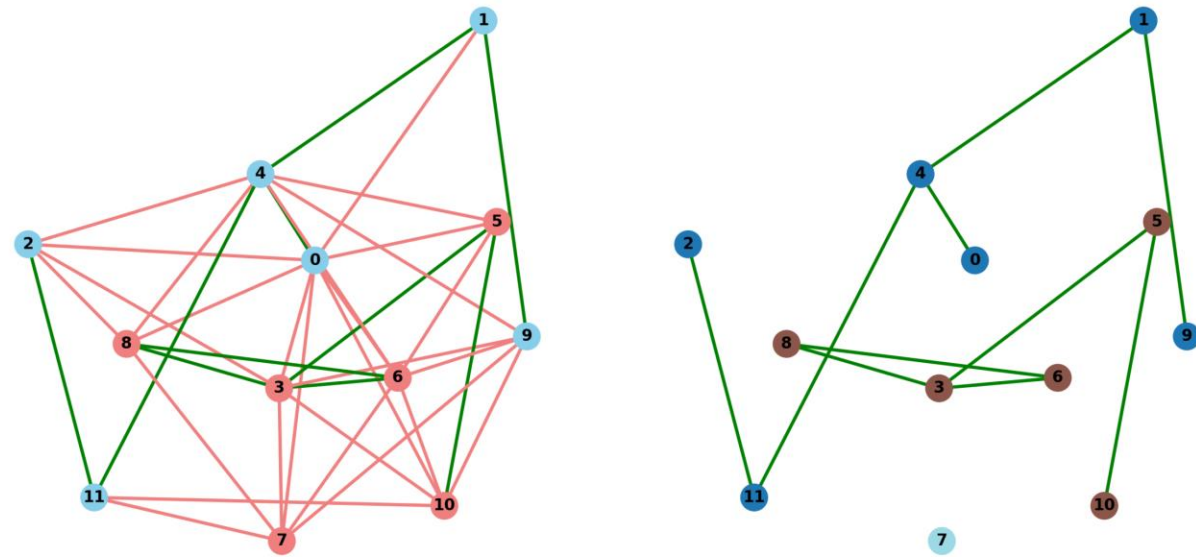
Cycle graph



Key Idea: Structure Determines Reliability

Edge-Cut

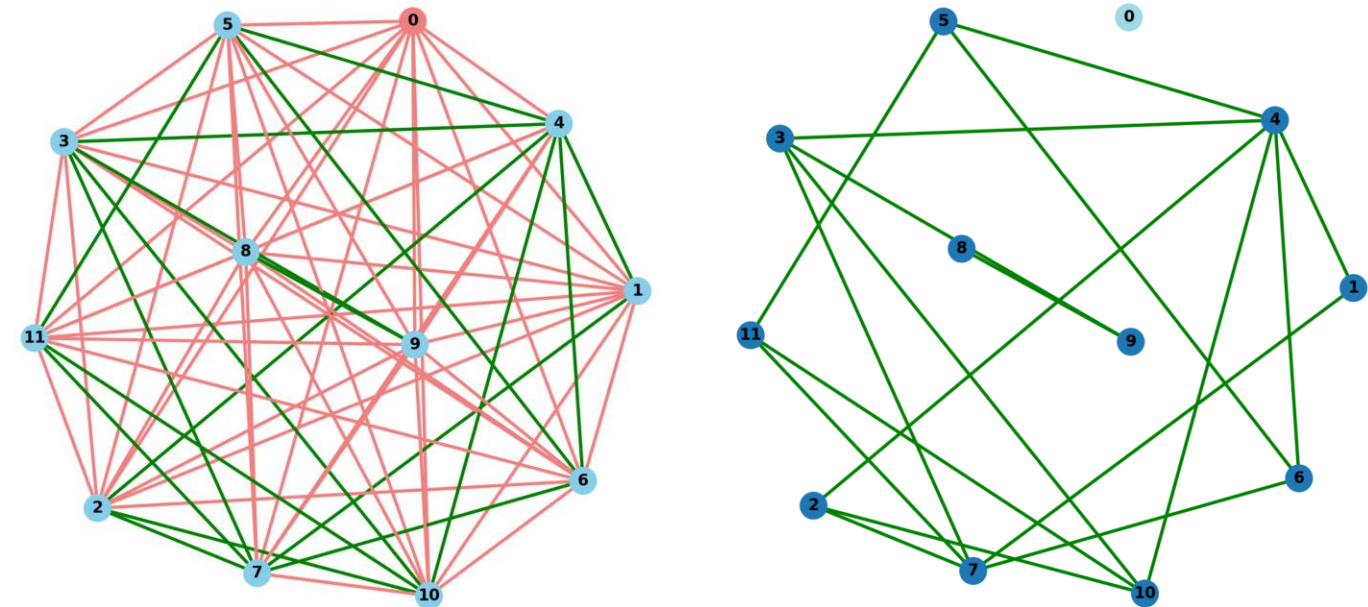
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Degree constrained graph
 $3 \leq \deg(v) < n-2$

Why It Matters

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Complete graph

Graph Theory Background

Cut-Edge (Bridge)

An edge whose removal disconnects the graph. If it fails, the network splits into two parts.

Edge-Cut

An edge cut is a set of edges whose removal immediately disconnects the graph into two or more components.

Minimum Cut — λ (lambda)

The smallest set of edges whose removal disconnects the graph. The larger λ is, the more reliable the network.

Monte Carlo Simulation Algorithm



Why Monte Carlo?

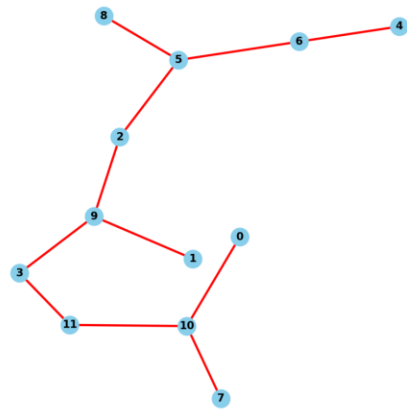
For large networks, calculating reliability exactly is very difficult. Monte Carlo provides a practical computational approach.

[Show Monte Carlo Demo](#)

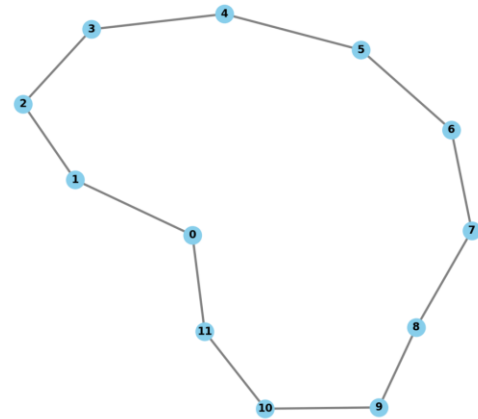
The Key Idea

1. Randomly fail edges with probability q
2. Check if the graph remains connected
3. Repeat thousands of times
4. Estimate reliability from results

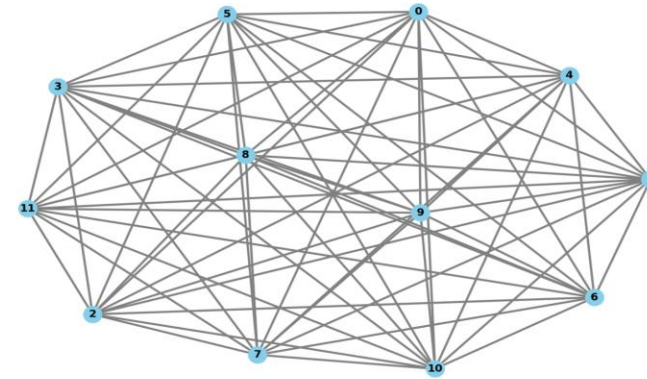
Four Graph Types Simulated



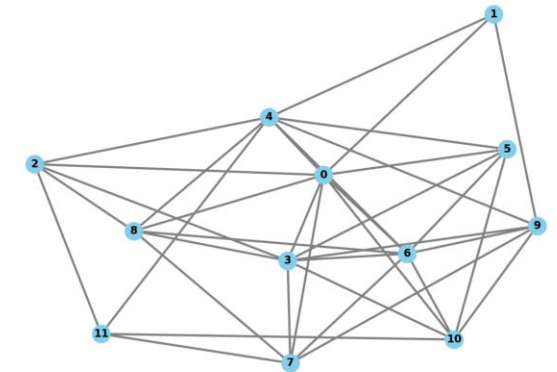
Tree Graph



Cycle Graph



Complete Graph



Degree-Constrained

$$3 \leq \deg(v) < n-2$$

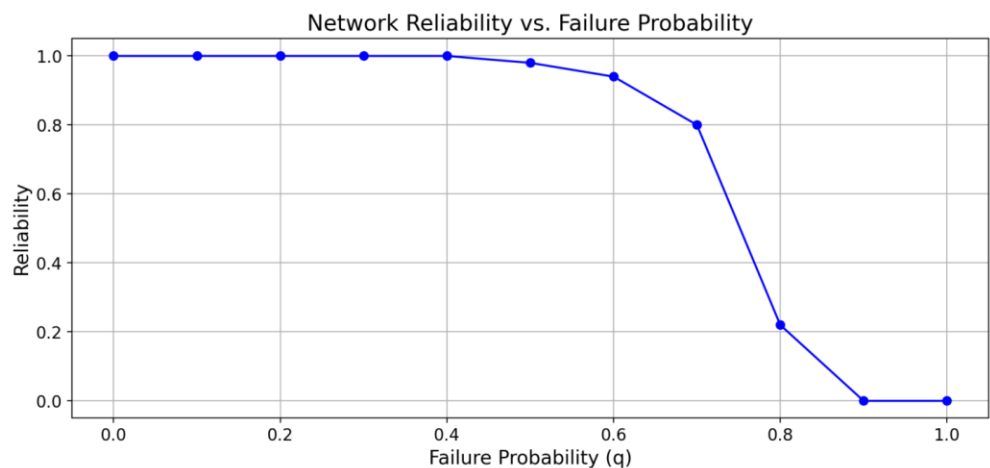
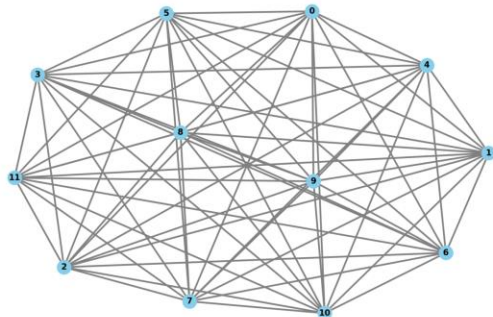
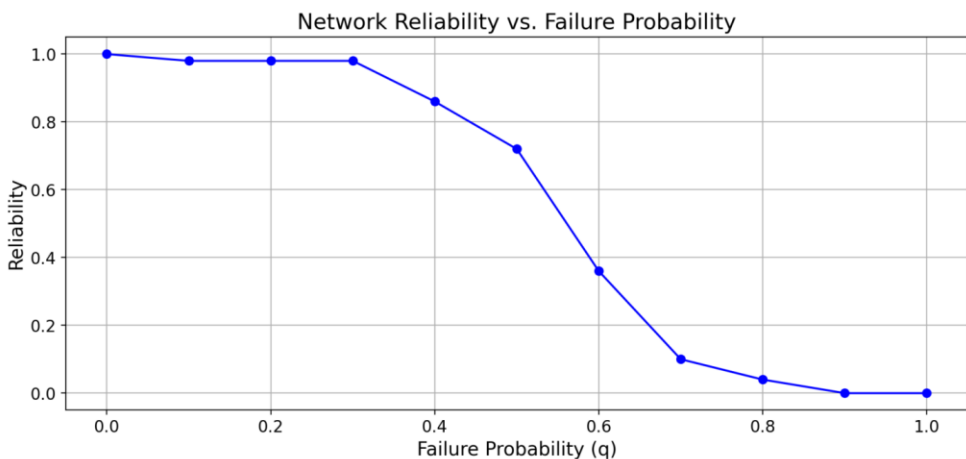
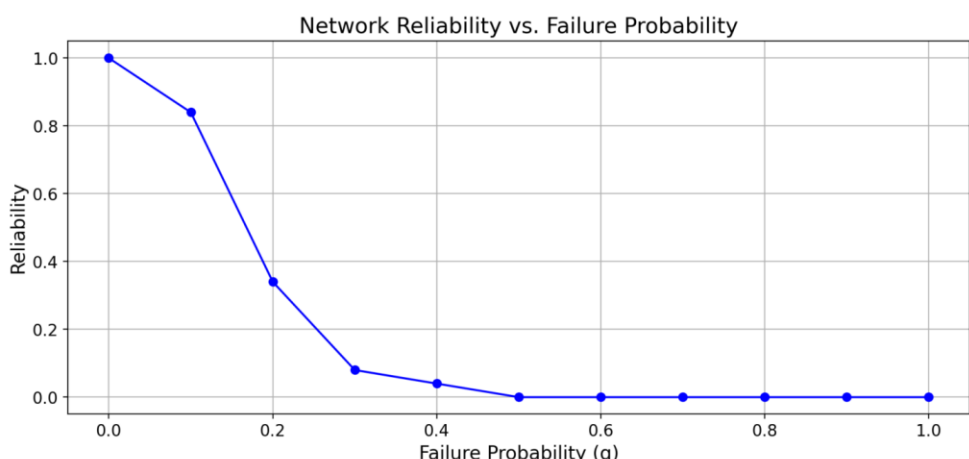
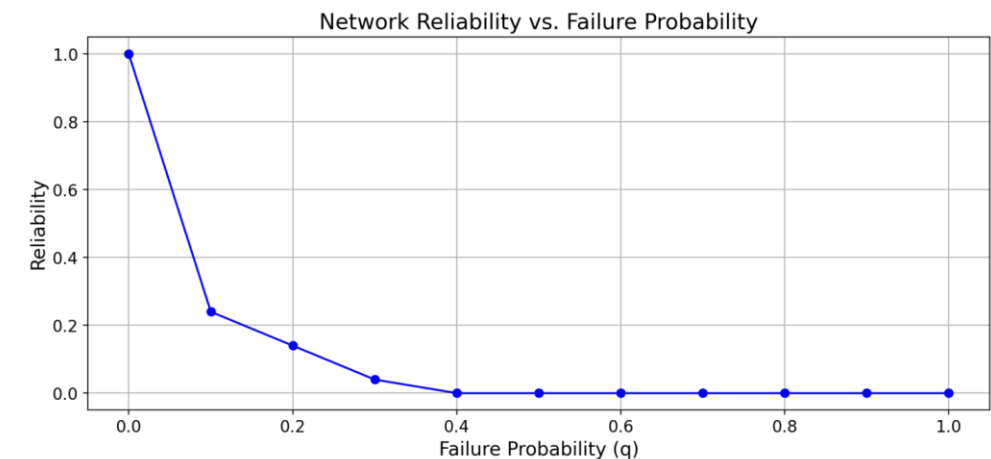
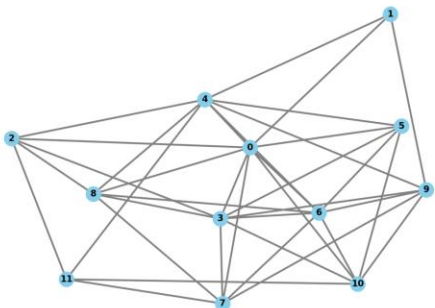
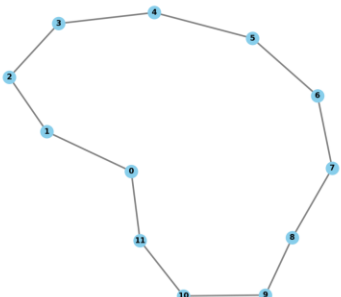
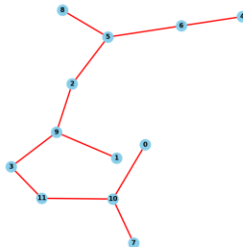
[Show Demo](#)

Results: Reliability vs. Failure Probability

N = 12

Q(probability) = 0.5

Monte Carlo Trials = 1200



Monte Carlo simulation confirmed all theoretical predictions across all four graph types.

Reliability: **connected trials ÷ total trials**

Conclusion

Structure Determines Reliability

Complete graphs are strongest; degree-constrained graphs reflect reality; trees are most fragile.

Monte Carlo is Effective

Simulation provides a practical method to estimate reliability for large, complex networks.

Network reliability depends on graph structure.

Thank You

Q & A